

Containerization with Docker and FastAPI project setup

Activity data

Activity number	1
Professor	Ramon Amela Milian
Teaching language	English
Association	Individual
Delivery date	22-06-2026

Activity description

In this activity, containerization principles will be worked on. The student will learn how to containerize and launch an HTTP service with all its dependencies. By building a docker container, the student will ensure that it is highly portable and easily deployable into any public cloud provider. The work will start from a minimalist FastAPI working project.

To minimize the time spent setting up the environment, instructions on how to virtualize an Ubuntu 22.04.4 LTS are provided in the virtual campus.

Competences

CE3 – Design and manage information systems for data treatment and storage

COMPETENCES (CE / CT)	LEARNING RESULTS (RA)	EVALUATION INDICATORS (IE)
CE3	RA1 – Summary the historical evolution and the current opportunities in the data system administration in cloud environments	IE1 – The student understands and uses modern virtualization techniques used in systems administration

	RA3 – Justify the use of cloud technologies to guarantee the scalability, accessibility, and coherence of a software solution	IE7 – The student can launch a custom application in a virtualized environment
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Activity contents

- Git
 - o Project initialization
 - o Use of a public cloud repository
- Docker
 - o Ability to add targets and modify them
- Docker compose
 - o Ability to modify existing targets with the most common modifications
 - o Ability to create new targets
- Project setup
 - o Ability to create new EP
 - o Ability to correctly use the router
 - o Ability to standardize the development process through code formatters

Task descriptions

To achieve the objectives presented in the description, the student will follow the next steps:

- Create GitHub accounts for each one of the project participants
- Create a private GitHub repository and initialize it with the code provided
- Invite the user @ramonamela as collaborator of the created project
- Copy the project template into the newly created project

- Make the first commit with the message “First commit”
- Add the needed requirements to the project
- Add as many targets in the Docker file as you think a scalable cloud project should have
- Modify the docker-compose file to use the previously created target
- Modify the docker-compose file to use an environment file
- Modify the docker-compose file to map the project port from 80 to the desired port (suggested ports are 8080 or 8000, but you can choose any other reasonable option).
- Include a target in the docker-compose to format the project’s code. This will be done through the following command:
 - o `entrypoint: sh -c "black --config .black . && ruff check --fix"`

Consider that, to execute the previous command, black version 24.1.0 and ruff version 0.1.14 should be installed.

- In the FastAPI project:
 - o Create two apps called “authentication” and “files”.
 - o Create routers inside both new applications
 - o Import the new routers from the main file
 - o Create the following endpoints in the newly created routers:
 - Authentication
 - register – POST
 - login – POST
 - logout – POST
 - introspect – GET
 - Files
 - files – GET
 - files – POST
 - files/{id} – GET
 - files/{id} – POST
 - files/{id} – DELETE
 - files/merge – POST

Note that these endpoints should appear in the swagger file and always return a 200. In this deliverable, no logic will be implemented in the endpoints.

The previous steps could be separated in the following summary points:

1. Git
2. Docker
3. Docker compose
4. Project setup

Formal aspects of tasks

All the project's code must be delivered in compressed format through the virtual campus. In addition, in the delivered text, the commit hash with the code in *GitHub* should be included. This hash should contain the same exact version as the delivered code.

Task	Percentage
1	25%
2	20%
3	25%
4	30%

Evaluation

Number	Task	Weight	IE
1	Containerization with Docker and FastAPI project setup	20%	IE1, IE7

Information resources

Capsules and videoconferences